MILITARY SPECIFICATION

CENTRAL OFFICE, TELEPHONE, MANUAL AN/MTC-1( )

1. SCOPE

1.1 Scope.- This document covers the following transportable shelter assemblages which comprise one type of a Telephone Central designated as Central Office, Telephone Manual AN/MTC-1( ): (See 6.5)

(1) ea Telephone Switchboard Group AN/MTA-3( )
(1) ea Telephone Terminal Group AN/MTA-4( )

The telephone central is capable of terminating 196 local and 20 trunk telephone lines.

2. APPLICABLE DOCUMENTS

2.1 Documents.- The following documents of the issue in effect on the date of invitation for bids, or request for proposal form a part of this document to the extent specified herein:

SPECIFICATIONS

FEDERAL

V-T-276	Thread, Cotton
V-T-295	Thread, Nylon
QQ-A-200	Aluminum Alloy Bar, Rod, Shapes and Tube, Extruded, General Specification
QQ-A-225	Aluminum Alloy Bar, Rod, Wire or Special Shapes, Rolled Drawn or Cold Finish, General Specification for
QQ-A-250	Aluminum Alloy Plate and Sheet; General Specification for
QQ-S-571	Solder, Tin Alloy and Lead Alloy
QQ-S-698	Steel, Sheet and Strip, Low Carbon
QQ-S-766	Steel, Plate, Sheet and Strip - Corrosion Resisting
QQ-T-830	Tube, Steel (Carbon, Mechanical, Seamless and Welded)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

TT-E-516	Enamel, Lusterless, Quick-drying, Styrenated Alkyd Type
TT-E-527	Enamel, Alkyd, Lusterless
TT-E-529	Enamel, Alkyd, Semi-Gloss

MILITARY

MIL-P-116	Preservation, Methods of
MIL-W-530	Webbing, Textile, Cotton, General Purpose, Natural or in colors
MIL-T-3530	Thread and Twine, Mildew Resistant or Water Repellent Treated
MIL-C-3885	Cable Assemblies and Cord Assemblies (for use in Electronic and Associated Electrical Equipment)
MIL-W-4088	Webbing, Textile, Woven Nylon
MIL-W-6858	Welding, Resistance, Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys, and Titanium, Spot and Seam
MIL-T-7928	Terminals, Lug and Splice, Crimp-Style Copper
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-F-14256	Flux, Soldering, Liquid (Rosen Base)
MIL-E-15090	Enamel, Equipment, Light-Gray Formula No. 111
MIL-W-27265	Webbing, Textile, Woven Nylon, Impregnated
MIL-E-46061	Enamel, Camouflage, Solar-and Heat-Reflecting
MIL-S-55505	Shelter, Electrical Equipment (Containing Equipment) Packaging and Packing of

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:18Z
Check the source to verify that this is the current version before use.

MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-1235	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ELECTRONICS COMMAND

DL-SC-A-539606	AACCMS Common items drawings for AN/MTC-1()

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol shall be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description.- Central Office, Telephone, Manual, AN/MTC-1() consists of Telephone Switchboard Group AN/MTA-3() and Telephone Terminal Group AN/MTA-4().

3.1.1 Telephone Switchboard Group AN/MTA-3().- The AN/MTA-3() consists of 1 each Shelter S-280()/G modified to include racks, brackstry, blower, air-conditioner, power and signal ducts, and hardware necessary to permit operation of the following installed major items:

Switchboard, Telephone, Manual SB-1398()/GTA-14(V); 3 each
Telephone Set TA-312()/PT; 2 each
Intercommunication Station IS-147()/FI; 1 each
Headset-Microphone H-210()/G; 9 each

3.1.2 Telephone Terminal Group AN/MTA-4().- The AN/MTA-4() consists of 1 each Shelter S-280()/G modified to include racks, brackstry, blowers, power and signal ducts, and hardware necessary to permit operation of the following installed major items:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:16Z
Check the source to verify that this is the current version before use.

Telephone Circuit, Line Relay TA-452()/GTA-14(V); 2 each
Telephone Circuit, Trunk Relay TA-453()/GTA-14(V); 1 each
Telephone Main Distribution Frame TA-454()/GTA-14(V); 3 each
Power Distribution Panel SB-1399()/GTA-14(V); 1 each
Telephone Set TA-312()/PT; 2 each
Intercommunication Station IS-147()/PI; 2 each

3.2 Preproduction sample.- Unless otherwise specified in the invitation for bids and contract, the contractor shall furnish one preproduction sample of the AN/MTC-1() for approval. (See 4.3)

3.3 Construction.-

3.3.1 Procurement Model.- A procurement model of the AN/MTC-1() will be available for inspection by prospective bidders and will be lent to the contractor. Unless otherwise specified herein or in the invitation for bid, physical construction of the equipment shall conform to the model.

3.3.2 Drawings.- The parts and assemblies listed on DL-SC-A-539606 shall be fabricated in accordance with the drawings designated therein and used in construction of the AN/MTC-1():

3.3.3 Operational equipment.- The Central Office, Telephone, Manual AN/GTA-14(V) is a major part of the AN/MTC-1() and will be Government furnished to the contractor. The contractor shall be responsible for any modifications to and the installation of the components of the AN/GTA-14(V) into the AN/MTA-3() and AN/MTA-4() in accordance with the model.

3.3.4 Materials.-

3.3.4.1 Fabric and thread.- Fabric and thread shall conform to the following:

a. The color of fabric shall be Olive-Drab No.7. The color of thread shall match shade S-1 of the United States Army Standard Color Card for Official Standardization Shades of Sewing Thread. Fabric and thread shall be preshrunk, or allowance shall be made for shrinkage, in order to provide for satisfactory fit of finished items both before and after they are immersed in water and then dried.

b. Thread shall be Cotton per V-T-276 and be given mildew resistant treatment in accordance with MIL-T-3530, Class II; and nylon per V-T-295.

c. Webbing shall be Cotton in accordance with MIL-W-530, Class 7; and nylon per MIL-W-4088 or Class R of MIL-W-27265.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

3.3.4.2 Metals.- Metals shall be selected from the appropriate detailed specification of the basic specification cited and as specified in the following:

a. Extruded aluminum shall be in accordance with QQ-A-200.

b. Structural aluminum shall be in accordance with QQ-A-225.

c. Sheet aluminum shall be in accordance with QQ-A-250. The floor plates in the assemblages shall be alloy 7075-T6.

d. Sheet and strip steel shall be in accordance with QQ-S-698. Equipment support channels shall be steel, CR composition 1020.

e. Sheet, strip, and plate steel, corrosion resistant shall be in accordance with QQ-S-766. Battery box and hydrometer holder components shall be Class 304, Condition A.

f. Steel tubing shall be in accordance with QQ-T-830.

3.4 Electrical installation.-

3.4.1 Power wiring AN/MTA-3().- The AN/MTA-3() power wiring shall be in accordance with the model and meet the following:

3.4.1.1 Insulation resistance.- Insulation resistance shall not be less than 100 megohms when tested in accordance with 4.6.1.1.

3.4.1.2 Operation of power distribution panel.- All components on all circuits of the distribution panels shall meet continuity and operational tests as follows:

Power Distribution Panel, Circuit Breakers No. 1 through No. 6 (See 4.6.1.3 through 4.6.1.7).

Emergency Light Test, (See 4.6.1.8)

3.4.2 Power wiring AN/MTA-4().- The AN/MTA-4() power wiring shall be in accordance with the model and meet the following:

3.4.2.1 Insulation resistance.- Insulation resistance shall not be less than 100 megohms when tested in accordance with 4.6.2.1.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

3.4.2.2 Operation of power distribution panel.- All components on all circuits of the distribution panels shall meet continuity and operational tests as follows:

Power Distribution Panel, Circuit Breakers No. 1 through No. 6 (See 4.6.2 through 4.6.2.7)

Emergency Light Test. (See 4.6.2.8)

3.4.3 Signal wiring, insulation resistance AN/MTA-3().- The signal wiring shall be in accordance with the model and the insulation resistance shall be not less than 100 megohms when tested in accordance with 4.6.3.1.

3.4.4 Signal wiring, insulation resistance AN/MTA-4().- The signal wiring shall be in accordance with the model and the insulation resistance shall be not less than 100 megohms when tested in accordance with 4.6.3.2.

3.4.5 Cable assemblies.- The following cable assemblies shall be fabricated in accordance with the model and meet MIL-C-3885. (See 4.4).

a. Cord assemblies for phones and intercoms.

b. Power cable assemblies for air-conditioner, power panel input, and battery interconnections.

c. Signal cable assemblies which connect with Signal Entrance Boxes, Signal Interconnecting Boxes, and Operational Central Office Components.

d. Cable assemblies, 20 cycle ringing and 48 VDC, which connect with Power Entrance Boxes and Operational Central Office Components.

3.4.6 Suppression of Radio Frequency interference.- The motors in the exhaust blowers and heaters, if other than as used in the model, shall meet the emanation limits of MIL-STD-461 for tactical equipment (See 4.4).

3.5 Systems operation test.- With all operating components of the AN/MTC-1() installed and connected within the shelters and all interconnecting cables between the AN/MTA-3() and AN/MTA-4() attached the equipment shall be capable of functioning in accordance with the following: (See 4.6.4).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

3.5.1 Power panel SB-1399()/GTA-14(V).- The power panel shall be capable of furnishing necessary voltages and ringing power for the switchboards. (See 4.6.4.2)

3.5.2 Subsidiary switchboard functions.- The following switchboard subsidiary functions shall be capable of being performed without fault:

Supervisory Recall Lamps (See 4.6.4.3)
Dial (See 4.6.4.4)
Hand Ringing Generator (See 4.6.4.6)
Non-Master Ringing (See 4.6.4.7)
Position Grouping (See 4.6.4.8)
Monitor (See 4.6.4.9)
Conference Call (See 4.6.4.10)

3.5.3 CB-LB line circuit operation.- With phones arranged as specified in paragraphs 4.6.4.5 and 4.6.4.11, communication through the switchboard shall be possible.

3.5.4 Information turret operation.- The information turret shall be capable of receiving and holding two subscriber phone calls at the same time. (See 4.6.4.13)

3.5.5 Trunk circuit test.- The equipment shall be capable of permitting two subscribers to communicate with each other through the switchboard. One subscriber shall represent a distant LB circuit. (See 4.6.4.12)

3.5.6 Spare lines.- With two telephones arranged for LB operation, communication shall be possible between signal connector No.9 the binding posts, the trunk MDF and the jack box. (See 4.6.4.16)

3.5.7 Intercommunication station test.- The Intercommunication stations shall be capable of transmitting and receiving. (See 4.6.4.14)

3.5.8 Supervisor's phones.- The phones at the desks of the Wire Chief and Chief Operator shall provide intercommunication when tested in accordance with 4.6.4.15.

3.6 Vehicular transportation.- The assemblages comprising the AN/MTC-1() shall be capable of being driven over cross-country terrain by military vehicles without sustaining any damage. (See 4.7 and 6.2(b)).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.7 Railroad transportation.- The assemblages comprising the AN/MTC-1( ) shall be capable of being loaded, blocked and braced on a flatcar for shipment by rail. When so loaded, the assemblages shall be subjected to and withstand the test of paragraph 4.8 without damage except as follows: Deformation in the ceiling and floor immediately above and below the switchboards, line and trunk relay frames, and main distribution frames shall not exceed 5/32-inch after completion of the tests.

3.8 Drops, flat and rotational.- The assemblages comprising the AN/MTC-1( ) shall withstand drops of 18 inches onto concrete. After each of the drop tests of paragraph 4.9.1 and 4.9.2 there shall be no damage to the assemblages except as follows: Deformation in the ceiling and floor immediately above and below the switchboards, line and trunk relay frames, and main distribution frames shall not exceed 3/32-inch after completion of all drops. Deformation of the battery acid tray shall not exceed 1/8-inch after completion of all drops. (See 6.2(b)).

3.9 Watertightness.- The assemblages comprising the AN/MTC-1( ) shall withstand the tests of paragraph 4.10. After the tests, there shall be no leakage into the assemblages, their walls, floors, roof or doors. (See 4.10)

3.10 Fording.- The assemblages comprising the AN/MTC-1( ) shall be capable of being immersed in water to a depth of 21 inches (measured from the bottom of the skids) for a period of 1 hour. Upon completion of the test there shall be no leakage of water into the assemblages or their walls or floors. (See 4.11)

3.11 Mounting inserts.- All mounting inserts shall be installed without use of auxiliary materials, such as adhesive, and when installed, each insert shall be capable of withstanding the axial and torque load referenced in paragraph 4.12. There shall be no damage to the insert or insert threads. The insert shall not have rotated from its original position, and there shall be no distortion of the shelter mounting members, the ceiling, wall or floor areas immediately surrounding the insert being tested. Testing of the inserts shall be accomplished before the inserts are used to fasten bracketry, racks, and the like.

3.12 Cleaning.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable portions of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be cleaned thoroughly and shall be free from particles of solder, flux, scale and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after assembly of the units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.13 Finish.- The assemblages comprising the AN/MTC-1() shall be finished in accordance with the following(see 6.3):

3.13.1 Exterior surfaces.- The exterior surfaces of facilities and/or assemblages such as covers of boxes, side flaps on covers, and shrouds for air-conditioners and blowers, will have the same finish as the exterior surfaces of the bare shelter (finish P515 per MIL-F-14072; then lusterless white enamel No 37875 per FED Spec TT-E-516, 0.75 to 1.0 mil dry film thickness; then final film of lusterless green enamel per MIL Spec MIL-E-46061 dry film thickness 1.5 to 2.0 mils).

3.13.2 Exterior hardware.- Exterior hardware for facilities and/or assemblages such as entrance boxes, latches, braces, underside of covers, etc, will be finished the same as exterior surfaces except less the white enamel undercoat. These components will be painted prior to assembly to the shelter and any touch up will be with the lusterless green enamel of the shelter.

3.13.3 Interior.- Finish per MIL-F-14072, P513. Follow with an application as below:

a. Ceiling, semi-gloss white enamel per Spec TT-E-529, color chip #27875 of Federal Standard No. 595.

b. Floor, lusterless gray enamel per Spec TT-E-527, color chip #36118 of Federal Standard No. 595.

c. Door and Door Brace, as in 3.13.1 above to match exterior.

d. Walls, semi-gloss light green enamel per TT-E-529, color chip #24533 of Federal Standard No. 595.

e. Bracketry, semi-gloss light gray enamel, per MIL-E-15090, Class 2. If bracketry is ferrous material and is not aluminum, finish per P213 of MIL-F-14072 instead of P513.

3.14 Marking.-

3.14.1 General.- Marking shall conform to Specification MIL-M-13231 and the model.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

3.14.2 Nameplates.- Each AN/MTA-3() and AN/MTA-4() assemblage comprising the AN/MTC-1() shall be provided with two nameplates in accordance with the model; i.e., one for the individual assemblage and one for the AN/MTC-1() system. Each nameplate shall be imprinted with a sequential serial number, and proper assigned nomenclature. (See 6.5)

3.14.3 Aircraft loading data plate.- Each assemblage comprising the AN/MTC-1() shall be provided a data plate in accordance with the model. The plate shall be imprinted with the dimensions which locate center-of-gravity in the three planes and the assemblage gross weight as determined by the contractor from the preproduction assemblages.

3.14.4 Interior and exterior marking.- All interior and exterior marking of the equipment shall be as dipicted on the model.

3.15 Riveting.- Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.16 Sealer.- All joints and raw edges or framing for power and signal entrance and interconnecting boxes, air-conditioner and exhaust blowers shall be sealed to prevent entrance of moisture to the core material of the shelter panels and to the interior of the shelter. All joints shall be sealed in a manner to prevent collection and retention of moisture. All exterior rivets and screws shall be dipped in sealer prior to installation. The sealer shall be one part Silicone Construction Sealant SE 1204 as made by General Electric Co. or equal.

3.17 Securing of parts.- Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain components of the AN/MTC-1() securely so that loosening, wear of mounting arrangements, or permanent separation of parts or components will not occur when the equipment is subjected to the specified transportation and drop tests.

3.18 Soldering.-

3.18.1 Solder and flux.- Soldering of electrical connections shall be made with composition SN60, Type S,R, or RMA in accordance with QQ-S-571. When type S is used, liquid rosin flux shall conform to MIL-F- 14256. No acid or acid salts shall be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.18.2 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loose.

3.19 Tropicalisation of material.- Canvas, webbing, wood, etc. shall be treated to be resistant to moisture and fungi as indicated in this specification as follows:

3.19.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.19.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.19.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.19.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use for materials such as canvas, webbing, cloth, etc. (See 6.2.(e)).

3.20 Welding.- Welded joints shall be as shown in the model and shall be such that grinding of the finished weld will be unnecessary. Spot, stitch, and seam welds shall conform to Specification MIL-W-6858, Class B. All surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from scale, paint, grease and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralised and removed. (See 4.4)

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:18Z
Check the source to verify that this is the current version before use.

3.21 Wiring and cabling.- Wiring and cabling shall be in accordance with the model and shall be neat and sturdy.

3.21.1 Slack.- Wires and cables shall be as short as practical except that slack shall be provided:

a. To prevent undue stress on cable forms, wires, and connections.

b. To facilitate field repair of broken or cut wires.

c. To prevent chafing or breaking of wires due to repeated flexing of hinged parts.

3.21.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.21.3 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.21.4 Crimping of solderless terminal lugs.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements Specification MIL-T-7928. (See 4.4 and 4.13)

3.21.5 Identification of wiring.- Wiring shall be identified by color coding and wire pair marking, as indicated by the model.

3.22 Interchangeability.- Like units, assemblies, sub-assemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. (See 4.14) Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristics, and the like.

3.23 Workmanship.- The AN/MTC-1() shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

(a) 3.3 Construction
(b) 3.12 Cleaning
(c) 3.13 Finish
(d) 3.14 Marking
(e) 3.15 Riveting
(f) 3.16 Seal, vapor
(g) 3.17 Securing or parts
(h) 3.18 Soldering
(i) 3.19 Tropicalisation of material
(j) 3.20 Welding
(k) 3.21 Wiring and cabling

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilise his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

a. Preproduction inspection (does not include preparation for delivery.) (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.16)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

4.3 Preproduction inspection.- This inspection will be performed by the contractor on the first article unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table I, the inspection specified in the subsidiary documents covering the items listed in para. 4.4, and the inspection specified for Group A, Group B, and Group C (see Tables II, III, and IV respectively). The preproduction inspection will normally be performed in this order: (a) Mounting inserts, (b) Power wiring, (c) Signal wiring, (d) System operation, (e) Vehicular transportation, (f) Rail transportation, (g) Drop tests, (h) Watertightness, (i) Fording. The last tests performed shall be a repeat of the System operation tests. The remaining preproduction tests shall be performed in any order satisfactory to the contracting officer.

Table I - Preproduction inspection

Inspection (For additional preproduction inspection see 4.3	Req't Para	Insp Para
Vehicular transportation	3.6	4.7
Rail transportation	3.7	4.8

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Cable assemblies	3.4.5
Suppression, RFI	3.4.6
Finish	3.13
Marking	3.14
Welding	3.20
Terminals	3.21.4

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specifications requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government-at its discretion- may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.7) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to Table II and the inspection procedures of Standard MIL-STD-105 using the general inspection levels. Group A inspection shall be performed in any order satisfactory to the Government.

Table II - Group A inspection

Inspection	Req't Para	Insp Para	AQL Major	AQL Minor
Visual and Mechanical	3.23	4.15	1.5 dpbw*	6.5 dpbw
Electrical			1% for the entire group of power and signal wiring and system operation	**
Power wiring AN/MTA-3()				
Insulation breakdown	3.4.1.1	4.6.1.1		
Distribution panel	3.4.1.2	4.6.1.3 thru 4.6.1.7		
Power wiring AN/MTA-4()				
Insulation breakdown	3.4.2.1	4.6.2.1		
Distribution panel	3.4.2.2	4.6.2.3 thru 4.6.2.7		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

Table II - Group A inspection (cont'd)

Signal wiring AN/MTA-3()	3.4.3	4.6.3.1
Signal wiring AN/MTA-4()	3.4.4	4.6.3.2
Systems operation AN/MTC-1()		
Power panel SB-1399()/GTA-14(V)	3.5.1	4.6.4.2
Subsidiary functions	3.5.2	4.6.4.3 4.6.4.4 4.6.4.6 thru 4.6.4.10
CB-LB line circuits	3.5.3	4.6.4.5 4.6.4.11
Information turret	3.5.4	4.6.4.13
Trunk circuit	3.5.5	4.6.4.12
Spare lines	3.5.6	4.6.4.16
Intercommunication stations	3.5.7	4.6.4.14
Supervisor's phones	3.5.8	4.6.4.15

* Defects Per Hundred Units (dphu)
** All defects are major

4.5.2 Group B inspection.- This inspection, including sampling, shall conform to Table III and to the procedures for small-sample inspection of Standard MIL-STD-105, using the special inspection levels. Group B inspection normally shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

4.5.2.1 Group B sampling plans.- The group B sampling plans, for the AQL's listed in Table III shall be as follows:

AQL	Inspection level
4.0%	S-4
6.5%	S-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.5.2.2 Order of inspection within Group B.- Group B inspection shall be performed on the same equipment in the order in which they appear.

Table III.- Group B inspection

Inspection	Req't Para	Insp Para	AQL (see 4.5.2.1)
Mounting insert	3.11	4.12	4.0%
Watertightness	3.9	4.10	6.5%
Fording	3.10	4.11	6.5%
Interchangeability	3.22	4.14	4.0%

4.5.3 Group C inspection.- This inspection shall be as listed in Table IV, and shall normally be performed on sample units that have been subjected to and met Group A and Group B inspection.

Table IV.- Group C inspection

Inspection	Req't Para	Insp Para
Drops, flat and rotational	3.8	4.9
Solderless-terminal-lug connections	3.21.4	4.13

4.5.3.1 Sampling for inspection of drop tests.- One sample shall be selected at random from the first 50 units or fraction thereof, and one sample from each additional 150 units or fraction thereof, produced.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

ilance or noncompliance with each requirement of the equipment specification shall be

4.5.3.2 Sampling for inspection of solderless-terminal-lug connections.- Five specimens of each combination of wire and terminal shall be prepared each week for inspection. A specimen is a 1-foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.3 Noncompliance.- The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and, (ii) all units of production which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.4 Reinspection of conforming Group B and Group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group B or Group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage.

4.5.5 Alternate inspection procedures.- Contractor may elect to use a continuous sampling and inspection plan instead of the lot inspection plan stated in 4.5.1 and 4.5.2. If this alternate is selected the procedure of implementation must be in accordance with MIL-STD-1235, "Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes". Before starting production Contractor shall notify the contracting office in writing which method he intends to use. The method may not be changed during production without written approval of the Government.

4.6 Electrical installation test.- The contractor shall provide all test equipment required for preproduction and production testing of the assemblages which comprise the AN/MTC-1(). Test equipment and any test fixtures proposed for use shall have approval of contracting officer prior to use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.6.1 Power wiring, AN/MTA-3(). Before applying power to the assemblage, an electrical ground at each Power Entrance Box shall be made. Power to the assemblages shall be 115 VAC, 60 hz, single phase.

4.6.1.1 Insulation resistance test.-

a Remove or disconnect all electrical loads:
Central office equipment
Fluorescent lamps and starters
Incandescent lamps
Neon lamps in Distribution Box
Power indicator neon lamp
Exhaust Blower, Air-conditioner & Wall Thermostat (Black Lead)
Heaters
Voltmeter in Distribution Box (isolate disconnected leads)
Fuse in Distribution Box

b. Place all breakers and switches in ON position

c. Connect a megohmmeter or equivalent tester between PIN 1 and PIN 2 of the Power IN connector in the Power Entrance Box. Adjust the tester for 500 volts DC. Maintain this potential for 1 minute or until the indicating meter has ceased fluctuating and indicates a steady or increasing insulation resistance. The requirement of 3.4.1.1 shall be met.

d. Repeat c. above with tester connected between PIN 1 and PIN 4; PIN 1 and shelter ground.

e. Repeat c. above with tester connected between pins of 48 VDC connector in Power Entrance Box; Shelter ground and pin (brass) counterclockwise from index key on shell of connector (Continuity shall exist between shelter ground and pin (silver) clockwise from index key on shell of connector).

f. Repeat c. above with tester connected between pins A and B of 20 cycle ringing connector in Power Entrance Box; Pins A-C, A-D-, B-C, B-D, and C-D; shelter ground and each pin.

g. Replace and reconnect all electrical loads.

4.6.1.2 Operation of power distribution panel.- With power connected, voltmeter on panel should read nominal 115 VAC. Neon lamp shall light indicating power is connected to shelter. Place Main Breaker in ON position.

4.6.1.3 Test of Circuit Breaker No.1.-

a. Place Circuit Breaker No.1 in ON position; neon lamp shall light. Place night light, fluorescent, and incandescent switches in their ON positions.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

b. Place the Blackout Bypass switch in the BYPASS position. All lamps shall light.

c. With shelter door open and the Blackout Bypass switch in the BLACKOUT position, all lights shall extinguish except the two blue lights over switchboards.

d. Close shelter door and note that all lights come on except the two blue lights which extinguish.

e. Open shelter door, place Blackout Bypass switch in the BYPASS position. All lights shall come on. Leave switch in this position for remainder of tests.

f. Operate Circuit Breaker No.1 to assure lights are controlled by this breaker. Leave breaker in the ON position.

g. Turn off Night Light and cold start incandescent switches.

4.6.1.4 Test of Circuit Breaker No.2 and No.3.-

a. Place Circuit Breaker No.2 in its ON position. Neon lamp shall light.

b. Place heater No.1 switch in the FAN position. Heater shall blow air.

c. Set thermostat to maximum temperature range.

d. Place switch in the HEAT position. After a short time the heater shall blow warm air.

e. Lower thermostat setting until a "click" is heard. This is the thermostat cutout and may be verified by observing a current drop in the power panel ammeter. The fan shall continue to operate.

f. With heater fan blowing, operate Circuit Breaker No.2 and verify it controls heater No.1. Leave breaker in the OFF position.

g. Repeat steps a.through f. for Circuit Breaker No.3 and heater No.2.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

4.6.1.5 Test of Circuit Breaker No.4.-

a. Place Circuit No.4 in its ON position. Neon lamp shall light.

b. Set Wall-Mounted Thermostat sufficiently low to permit sensing a higher ambient temperature.

c. Adjust Air-Conditioner Thermostat to MAXIMUM COOL position. Place Function switch in FAN position. Fan shall blow air.

d. Place Function switch in COOL position. Compressor shall come on. This can be verified by observing an increased current reading at the power panel ammeter.

e. After a short time the air conditioner shall blow cold air.

f. Increase set point of wall-mounted thermostat to a temperature higher than inside shelter temperature. Air-Conditioner shall stop.

g. With air-conditioner fan running, operate Circuit Breaker No.4 and verify it controls air-conditioner. Leave breaker in the OFF position.

4.6.1.6 Test of Circuit Breaker No.5.-

a. Place circuit breaker No.5 in its ON position. Neon lamp shall light.

b. With a test lamp plugged into the SB-139()/GTA-14(V) outlet, operate the current breaker and verify it controls this outlet.

c. Remove test lamp and leave circuit breaker in its OFF position.

4.6.1.7 Test of Circuit Breaker No.6.-

a. Place Circuit Breaker No.6 in its ON position. Neon lamp shall light.

b. Open Exhaust Blower vent cover located on bottom of blower housing on exterior of shelter.

c. Place Exhaust Blower switch in its ON position. Blower shall exhaust shelter air.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

d. Operate circuit breaker and verify it controls blower.

e. Place Exhaust Blower switch in its OFF position.

f. Plug in Intercom power cord in the Intercom outlet. Turn Intercom on and note that power ON lamp lights.

g. Operate circuit breaker and verify it controls Intercom.

h. Plug a test lamp into Convenience Outlets (curbside wall (2) each) and (roadside wall (1) each). Test lamp shall light in each case.

i. Operate Circuit Breaker No.6 and verify it controls these outlets.

j. Repeat this procedure for duplex outlet in Power Entrance Box. Leave breaker in its OFF position.

4.6.1.8 Emergency light test.-

a. Simulate the installation of 48v batteries (normally provided from the AN/MTA-4) by providing a 48 VDC source to the emergency light circuit.

b. Place emergency light circuit breaker located in raceway, in its ON position.

c. Throw main breaker (115 VAC) to OFF (simulates loss of power). All lights shall extinguish and the emergency lights shall come on.

d. Place emergency light circuit breaker in its OFF position and throw main breaker to ON. Emergency lights shall extinguish and fluorescent lights come on.

4.6.2 Power wiring AN/MTA-4().- Before applying power to the assemblage an electrical ground at each Power Entrance Box shall be made. Power to the assemblages shall be 115 VAC, 60 hz, single phase.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.6.2.1 Insulation resistance test.-

a. Remove or disconnect all electrical loads:

Central Office Equipment
Fluorescent lamps and starters
Incandescent lamps
Neon lamps in Distribution Box
Power indicator neon lamp
Exhaust blowers
Heaters
Voltmeter in Distribution Box (isolate disconnected leads)
Fuse in Distribution Box

b. Place all breakers and switches in ON position.

c. Connect a megohmmeter or equivalent tester between Pin 1 and Pin 2 of the Power IN connector in the Power Entrance Box. Adjust the tester for 500 volts DC. Maintain this potential for 1 minute or until the indicating meter has ceased fluctuating and indicates a steady or increasing insulation resistance. The requirement of 3.4.2.1 shall be met.

d. Repeat c above with tester connected between Pin 1 and Pin 4; Pin 1 and shelter ground.

e. Repeat c above with tester connected between Pins of 48V DC connector in Power Entrance Box; Shelter ground and terminal clockwise from index key on shell of connector (continuity shall exist between shelter ground and the other terminal in the connector).

f. Repeat c above with tester connected between Pins A and B of 20 cycle ringing connector in Power Entrance Box; Pins A-C, A-D, B-C, B-D, and C-D; shelter ground and each pin.

g. Replace and reconnect all electrical loads.

4.6.2.2 Operation of power distribution panel.- With power connected, voltmeter on panel should read nominal 115 V AC. Neon lamp shall light indicating power is connected to shelter. Place Main Breaker in ON position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.6.2.3 Test of Circuit Breaker No.1.-

a. Place Circuit Breaker No.1 in ON position; Neon lamp shall light. Place night light, fluorescent, and incandescent switches in their ON positions.

b. Place the Blackout Bypass switch in the BYPASS position. All lights shall come on.

c. With shelter door open and the Blackout Bypass switch in the BLACKOUT position all lights shall extinguish except the two blue lights over distribution frames.

d. Close shelter door and note that all lights come on except the two blue lights which extinguish.

e. Open shelter door, place Blackout Bypass switch in the BYPASS position. All lights shall come on. Leave switch in this position for remainder of tests.

f. Operate Circuit Breaker No.1 to assure lights are controlled by this breaker. Leave breaker in the ON position.

g. Turn off Night Light and cold start incandescent switches.

4.6.2.4 Test of Circuit Breaker No.2 and No.3.-

a. Place Circuit Breaker No.2 in its ON position. Neon lamp shall light.

b. Place heater No.1 switch in the FAN position. Heater shall blow air.

c. Set thermostat to maximum temperature range.

d. Place switch in the HEAT position. After a short time the heater shall blow warm air.

e. Lower thermostat setting until a "click" is heard. This is the thermostat cutout and may be verified by observing a current drop in the power panel ammeter. The fan shall continue to operate.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the currentversion before use.

f With heater fan blowing, operate Circuit Breaker No.2 and verify it controls heater No.1. Leave breaker in the OFF position.

g Repeat steps a. through f. for Circuit Breaker No.3 and heater No.2.

4.6.2.5 Test of Circuit Breaker No.4.-

a. Place Circuit Breaker No.4 in its ON position. Neon lamp shall light.

b. Open exhaust blower vent covers on bottom of blower housings on exterior of shelter.

c. Place exhaust blower switch No.1 in its ON position. Blower shall exhaust shelter air.

d. Operate circuit breaker to verify it controls the blower.

e. Place switch No.1 in its OFF position.

f. Repeat a. through e. above for blower No. 2.

4.6.2.6 Test of Circuit Breaker No.5.-

a. Place Circuit Breaker No.5 in its ON position. Neon lamp shall light.

b. With a test lamp plugged into the SB-1399()/GTA-14(V) outlet in raceway, operate the circuit breaker and verify it controls this outlet.

c. Remove test lamp.

d. Remove battery exhaust plug from battery exhaust vent on the exterior of the shelter.

e. Plug battery exhaust blower power cord into battery exhaust outlet. Observe that air is being exhausted through the vent to the exterior of the shelter.

f. Operate the circuit breaker and verify it controls the outlet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

g. Leave circuit breaker in its OFF position.

4.6.2.7 Test of Circuit Breaker No.6.-

a. Place Circuit Breaker No.6 in its ON position. Neon lamp shall light.

b. Plug in the two intercom power cords into their power outlets. Turn intercoms on and note that their power ON lamps light.

c. Operate circuit breaker and verify it controls intercom outlets.

d. Turn intercoms to their OFF positions.

e. Plug a test lamp into each convenience outlet (2 roadside, 2 curbside) and note that lamp lights.

f. Operate Circuit Breaker No.6 and verify it controls these outlets.

g. Repeat this procedure for duplex outlet in Power Entrance Box. Leave breaker in its OFF position.

4.6.2.8 Emergency light test.-

a. Install 48V batteries or their equivalent in battery box and connect to Cable from Power Panel.

b. Place Emergency Light circuit breaker located in conduit between Power Distribution Panel and SB-1399()/GTA-14 in its ON position.

c. Throw main breaker to OFF (Simulates loss of 115 VAC power). All lights shall extinguish and the emergency lights shall come on.

d. Place Emergency Light Switch Circuit Breaker in its OFF position. Throw main breaker ON. Emergency lights shall remain extinguished. Leave main breaker in its ON position.

4.6.3 Signal wiring.-

4.6.3.1 Insulation resistance test AN/AIA-3().-

a. Remove all plugs from the Phone 1, Phone 2, Local Intercom, and Info Phone jacks.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

b. Connect a megohmmeter or equivalent tester between first pair of BINDING POSTS in Signal Interconnecting Box A. Adjust the tester for 500 volts DC. Maintain this potential for 1 minute or until the indicating meter has ceased fluctuating and indicates a steady or increasing insulation resistance. The requirement of 3.4.3 shall be met.

c. Repeat b. above with tester between remaining pairs of BINDING POSTS; between each BINDING POST and shelter ground.

NOTE: To Eliminate Danger of Shock.- Discharge conductors following each test by connecting a 500 ohm wirewound resistor, or equivalent test lamp to the pins tested.

4.6.3.2 Insulation breakdown test AN/MTA-4().-

a. Remove all Clinch-Type connectors from the switchboard side of the Main Distribution Frames.

b. Remove all plugs from the Spares, Intercoms, Phone 1, and Phone 2 jacks at the supervisor's desk.

c. Connect a megohmmeter or equivalent tester between first pair of BINDING POSTS in Signal Interconnecting Box A. Adjust the tester for 500 volts DC. Maintain this potential for 1 minute or until the indicating meter has ceased fluctuating and indicates a steady or increasing insulation resistance. The requirement of 3.4.4 shall be met.

d. Repeat c. above with tester between remaining pairs of BINDING POSTS; between each BINDING POST and shelter ground.

e. Using a low resistance continuity tester measure continuity between PIN 1A of Signal 1 Connector in Signal Entrance Box C and Upper BINDING POST of Line 00 pair; PIN 1B and Lower Binding Post of Line 00 pair; thru PIN 26B and Lower Binding Post of Area Intercom pair. Repeat procedure for PINS and BINDING POSTS for Signal 2 thru 4 in Box C; Signal 5 thru 8 in Box D; and Signal 9 in Box E.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

f. Repeat c. above with the tester between PIN 1A of Signal 1 connector and shelter ground; between each remaining pin of all Signal 1 thru 9 connectors and shelter ground; between each pin of Signal 1 connector and each remaining PIN within that connector and similarly for Signal 2 thru 9 connectors.

g. Repeat c. above with the tester between Info 1 BINDING POSTS in Signal Entrance Box C; Info 2 BINDING POSTS in Box C; each Info BINDING POST and shelter ground.

NOTE: To eliminate danger of shock.- Discharge conductors following each test by connecting a 500 ohm wirewound resistor, or equivalent test lamp to the pins tested.

4.6.4 Systems Operation Test.-

4.6.4.1 Shelter connections and interconnections.- Connect the 25 ft long CX-7453()/U to the Power Out connector in the Power Entrance Box of the AN/MTA-4() and the Power IN connector in the Power Entrance Box of the AN/MTA-3(). Connect a 115 volt AC, 60 hz, single phase source to the Power IN connector in the Power Entrance Box of the AN/MTA-4(). Place main breaker, circuit breaker #1 (lights) and light switches in each assemblage in ON position. The equipment circuit breakers #5 in each Power Distribution Panel and all Breakers on Power Panel in the AN/MTA-4() shall be in OFF position. Connections within and between the shelter assemblages shall be as follows:

a. In the AN/MTA-4():

(1) Line pairs from Signal Entrance Boxes shall terminate to BINDING POSTS on Line side of MDFs.

(2) Jumpers on Switchboard side of MDFs between Line and Switchboard BINDING POST sets shall be in place.

(3) Clinch-type connectors on Telephone cables shall be connected to corresponding connectors of MDFs and Relay Racks.

(4) Clinch-type connector on cables from Interconnecting Boxes shall be connected to associated Relay Rack connectors.

(5) Connect 48 volt cables between Power Panel and Relay Racks.

(6) Connect 20 Cycle Ringing and 48 volt connectors (for Switchboard use) in the Power Panel with associated connectors on cables from Power Entrance Box.

(7) Connect the Night Alarm cable terminations to terminals on the Relay Racks.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

(8) Connect 48 volt connector from Battery Box cable to the Power Panel.

(9) Connect 110 volt Power Input cable between the Power Panel and the Raceway.

b. In the AN/MTA-3():

(1) Clinch-type connectors on telephone cables from Interconnecting Boxes shall be connected to corresponding connectors on rear of Switchboard 1.

(2) Clinch-type connectors on telephone cables shall be connected to corresponding connectors of Switchboards 1 and 2; and Switchboards 2 and 3.

(3) Connect 48 volt cables: from Power Entrance Box to Switchboard 1; and between Switchboards.

(4) Connect 20 Cycle Ringing cables: from Power Entrance Box to Switchboard 1; and between Switchboards.

(5) Connect 110 volt cables: from Roadside Raceway to Switchboard 1; and between Switchboards.

(6) Connect Grouping Cables between Switchboards.

c. Interconnections from AN/MTA-4() to AN/MTA-3():

(1) Connect 22 telephone cables, 25 ft long CX-4566()/G, to corresponding connectors in the Interconnecting Boxes of the assemblages.

(2) Connect the 25 ft long Information Cable to connector in the Interconnecting Box A of each assemblage.

(3) Connect the 25 ft long 48 volt, and 20 Cycle Ringing-Night Alarm cables to connectors in the Power Entrance Box of each assemblage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.6.4.2 Power panel test, SB-1399()/GTA-14(V).-

a. Place 48V breaker on top panel to ON position. There shall be no indication of alarm, buzzers, etc., within the shelter.

b. Place 110 VAC breaker on top panel to ON position.

c. Turn FINE and COURSE knobs on lower panel to their lowest settings.

d. Place AC INPUT breaker on lower panel to ON position. Pilot light shall come on.

e. Place the DC OUTPUT breaker on lower breaker to ON position.

f. Adjust the COURSE and FINE knobs until 48 volts is obtained on the meter.

g. Place TELERING breaker & TELERING switch on top panel to ON position.

h. Measure for approximately 48 VDC between any fuse and ground at each trunk and line relay rack. Note that ground is positive.

i. Carefully depress the tabs of each fuse on the Line Relay Rack and Trunk Relay Rack. Buzzer alarm shall sound when each tab is depressed.

j. Measure for approximately 48 VDC between any fuse and ground at the switchboards.

k. Measure for approximately 100 VAC, ± 5volts, at the 20 cycle ringing plugs connected to each switchboard.

l. Measure for approximately 115 VAC at the AC plugs connected to each switchboard.

m. Verify that each heater in switchboards is heating.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.6.4.3 Supervisory recall lamps test.-

a. Place ANS cord plug half-way into "recall" jack (lower right hand corner on face equipment). Supervisory lamp shall light.

b. Insert ANS cord plug completely into "recall" jack. Supervisory lamp shall extinguish.

c. Repeat these two steps for all ANS and CALL cords on each Switchboard. All lamps shall function.

4.6.4.4 Dial test.-

a. Place first RING-ANSWER-DIAL switch, starting from left of No. 1 switchboard, in its locked position. (No. 1 switchboard is unit nearest front wall; No.3 unit is nearest door end).

b. With an ohmmeter set for R x 1 scale reading, place meter leads between sleeve of first (from left) CALL cord plug and Ground on key shelf.

c. Dial Zero and hold the dial at this position. Ohmmeter shall read between 2 and 5 ohms.

d. Release dial. When dial reaches zero, ohmmeter shall read infinity.

e. Return switch to its normal position.

f. Repeat steps a. through e. for all 15 plugs.

g. Repeat steps a. through f. for Switchboard No. 2 and No. 3.

h. With ohmmeter set for R x 10 scale reading, place meter leads between tip and ring of first (from left) CALL cord plug of No. 1 Switchboard.

i. Place RING-ANS-DIAL switch in its locked position. A reading of 500 ohms ± 10% shall be obtained.

j. With leads still across tip and ring, dial zero and hold dial at this position. A reading of 5 ohms ± 10% shall be obtained.

k. Release dial and return switch to normal position.

l. Repeat steps h through k for all 15 plugs.

m. Repeat steps h through l for Switchboard No. 2 and No. 3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

n. With ohmmeter set for R x 10 scale, place meter leads between tip and ring of any CALL cord plug on No. 1 Switchboard. Throw DIAL-WIPEOUT switch to WIPEOUT position. A reading of infinity shall be obtained.

o. With ohmmeter set for R x 1 scale, place meter leads across sleeve contact of any CALL cord plug and Ground on Key Shelf. A reading of between 2 and 5 ohms shall be obtained when DIAL-WIPEOUT switch is thrown to the WIPEOUT position.

p. Repeat steps n and o for Switchboard No. 2 and No. 3.

4.6.4.5 CB line circuit test.-

a. Place all CB-LB switches on each line relay rack in their CB position.

b. Place MA switch in the ON position at each relay rack.

c. Connect a CX-4566()/G cable between one Distribution Box J-1077()/U and Signal Connector No. 1 in the Signal Entrance Box "C". Make this same cable hookup using the remaining Distribution Box J-1077()/U and Signal Connector No. 5 in Signal Entrance Box "D".

d. Connect a TA-312()/PT phone arranged for CB operation to BINDING POST pair No. 1 of one J-1077()/U and another TA-312()/PT phone arranged for CB operation to pair No. 1 of the other J-1077()/U. These phones will be referred to as Phone No. 1 and No. 2 respectively.

e. Connect a headset to one of the headset receptacles on the face of the keyshelf at operator's position No. 1, (nearest front wall). Connect a headset to operator's position No. 2 and No. 3 in a similar manner.

f. Lift Phone No.1. Line Lamp 00 on face equipment shall light. The night alarm shall ring. NOTE: The night alarm may be muted since repeated operation will occur during test of all lines.

g. Insert ANS cord into jack associated with the lighted line lamp at operator's position No.1. The lamp shall extinguish and the Night Alarm shall stop ringing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

h. Place RING-CALL-TALK switch in the TALK (locked) position. Communication through Phone No.1 and operator's headset shall be loud and clear.

i. Repeat sequence of steps f.,g., and h. above for Phone No.2 using CALL cord. Communication shall be loud and clear between Phones No.1 and No.2.

j. Replace Phone No.1 and No.2 in their cradles. The CALL SUPERVISORY LAMP and ANS SUPERVISORY LAMP shall light.

k. Momentarily operate both RING-CALL-TALK and RING-ANS-DIAL switches to their RING positions. Phone No.1 and No. 2 shall ring respectively.

l. Withdraw both plugs from the jacks in the face equipment. All lamps shall extinguish.

m. Repeat steps f. through l. at operator's position No.2 and No.3.

n. Repeat steps f. through m. for each pair 2A and 2B thru 25A and 25B. Use all ANS and CALL cords (15) at each operator's position until each is verified as satisfactory.

o. After completion of test for Signal Connector No.1 and No.5, move cables to Signal Connector No.2 and 6, No. 3 and 7, and No. 4 and 8 respectively and repeat steps f. thru n.with the following exceptions:

(1) Use of all ANS and CALL cords is not required.

(2) At any convenient time, substitute the remaining six (6) operator's headsets for verification of their satisfactory operation.

(3) At any convenient time, utilize the remaining three (3) headset receptacles on the face of the key shelf.

(4) At any convenient time, temporarily place EMERGENCY RING BREAKER in ON position and TELERING EMERGENCY RING switch in the EMERGENCY RING position. A complete operation sequence of the two phones shall be possible. Return breaker and switch to previous positions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

p. Leave all connections unchanged after completing test.

4.6.4.6 Hand ringing generator test.-

a. Place the RINGING-KEY-HAND switch on Switchboard No.1, face equipment to the HAND position.

b. Lift handset of Phone No.1 momentarily to assist operator in locating of jack and insert CALL cord plug into jack on face equipment that lights.

c. Place the RING-CALL-TALK switch in the RING position.

d. Turn hand crank on the switchboard shelf. Phone No.1 shall ring. Return RINGING-KEY-HAND switch to KEY position and remove plug.

e. Repeat steps a. through d. for Switchboard No.2 and No.3. Leave all connections unchanged after completing test.

4.6.4.7 Monitor-master ringing test.-

a. Place an AC voltmeter (with any scale between 0 to 150 and 0-1000) across the RING of a CALL cord plug on Switchboard No.1 and a ground on the key shelf.

b. Place RING-CALL-TALK switch in RING position (non-lock); read approximately 100 volts on the meter. Release switch.

c. Place the voltmeter across tip of the CALL cord plug and a ground on the key shelf.

d. Place RING-CALL-TALK switch in RING position; read 0 volts on meter.

e. Place MONITOR-MASTER-RING switch (red handle on right) in locked position; read 100 volts.

f. Place meter across ring of CALL cord plug and a ground on the keyshelf; read 0 volts. Return switches to normal.

g. Repeat a. through f. for Switchboard No.2 and No.3.

be sprayed ten equipment shall be subjected to the test

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.

shall all

4.6.4.8 Position grouping test.-

a. Lift handset of Phone No.2. Lamp at jack on face equipment shall light.

b. Insert ANS cord plug into this jack at Switchboard No.3. Lamp shall extinguish.

c. Place POSITION-GROUPING switch (red handle on left) of Switchboard No.3 in its locked position.

d. Communication between Phone No.2 and Switchboard No.2 operator shall be loud and clear.

e. Place POSITION-GROUPING switch of Switchboard No.2 in its locked position.

f. Communication between Phone No.2 and Switchboard No.1 operator shall be loud and clear.

g. Remove cord plug and return switches to normal position. Leave all connections unchanged after completing test.

4.6.4.9 Monitor test.-

a. Lift handset of Phone No.1. Lamp at jack on face equipment shall light.

b. Insert ANS cord plug into this jack at Switchboard No.1.

c. Place RING-CALL-TALK switch to the TALK position.

d. Phone No.1 operator and Switchboard operator shall be in communication.

e. Operator at switchboard shall give a long count. Phone No.1 operator shall not be able to hear long count when MONITOR-MASTER-RINGING switch (red handle on right) is placed in the MONITOR position (non-lock) by the Switchboard operator. The Switchboard operator shall be able to hear Phone No.1 operator.

f. Remove ANS cord plug and return switches to normal position.

g. Repeat steps a. through f. for Switchboard No.2 and No.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.6.4.10 Conference call test.-

a. Repeat 4.6.4.9 a. through d.

b. Place CALL cord plug into first conference jack (lowest row, right half of face equipment) at Switchboard No.1.

c. Insert second ANS cord plug into second conference jack.

d. Lift handset of Phone No.2 and insert CALL cord plug associated with second ANS cord into jack on face equipment that lights. Place second RING-CALL-TALK switch in TALK position; communication between switchboard and phones and between phones shall be loud and clear.

e. Place RING-CALL-TALK switches in normal position. Communication between phones still exists.

f. Remove plug from second conference jack and place in third conference jack. Communication between phones still exists. Place RING-CALL-TALK switch in TALK position; communication between switchboard and phones exists.

g. Repeat repositioning of latter cord plug until all 19 conference jacks are verified for loud and clear communications.

h. Repeat steps a through g for Switchboard No.2 and No.3. When completed, remove cord plugs.

4.6.4.11 LB line circuit test.-

a. Place all CB-LB switches on the line relay racks in their LB positions.

b. Arrange Phone No.1 and Phone No.2 for LB operation.

c. Perform the same sequence of operations as outlined in paragraph 4.6.4.5 with the following differences:

(1) Both phones will require hand ringing to signal switchboard and for ringdown when calls are completed.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the currentversion before use.

(2) Magneto supervisory lamps will light in lieu of supervisory ANS and CALL lamps when ringdown is accomplished at Phone No.1 and No.2.

(3) Leave phones arranged for LB operation.

4.6.4.12 Trunk circuits tests.-

a. Connect the CX-4566()/G cable from J-1077()/U Interconnecting Box with Phone No.1 to Signal Connector No.9. Leave remaining J-1077()/U hook-up as is.

b. Operate hand ringer of the Phone No.2. Lamp on face equipment shall light.

c. Insert ANS cord into line jack. Lamp shall extinguish. Communication between phone and operator shall be loud and clear.

d. Insert CALL cord into first trunk line jack.

e. Lift Phone No.1

f. Communication between Phone No.1 and No.2 shall be loud and clear.

g. Repeat steps b. through f. for each Trunk line (20).

h. Repeat steps b. through g. for remaining two switchboards.

4.6.4.13 Information turret (Info phones) test.-

a. Using a low resistance continuity tester measure continuity between corresponding BINDING POSTS marked INFO 1 in Signal Interconnecting Box "A" in the AN/MTA-3() and in Signal Entrance Boxes "C", "D" and "E" in the AN/MTA-4().

b. Repeat a. above for BINDING POSTS marked INFO 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

c. Place wire jumpers between: Line 96 on the Trunk MDF and Line MDF No.1; and similarly for Line 97

d. Connect a TA-312()/PT phone with a telephone cord to INFO phone jack over the INFO TURRET in the AN/MTA-3().

e. Originate a call from Phone No.1 on pair 1A and 1B of the J-1077()/U connected with No.9 Signal Connector (Signal Entrance Box "E" in the AN/MTA-4().)

f. Switchboard operator accepts call with ANS cord plug.

g. Switchboard operator inserts associated CALL cord plug into Information Line No.1 jack on face equipment (corresponds to line 96).

h. Switchboard operator rings (RING-CALL-TALK switch to RING position) Information Turret position No.1 at supervisor's desk. CALL light on information turret shall blink when operator rings position.

i. Place Information Turret Switch in TALK position. Establish communication between phone outside shelter and Info Phone 1 at desk.

j. Place Info Turret Switch Line No.1 in HOLD position. Phone No.1 and Supervisor's position shall not be capable of communicating. HOLD lamp shall light.

k. Originate a call from a second phone connected to the same J-1077()/U.

l. Switchboard operator places Phone No.2 through to Information Line No. 2 (corresponds to Line 97 on Switchboard). Phone No.1 shall not hear conversation.

m. Place Information Line No.2 in HOLD position. HOLD lamp shall light. Operator at desk and Phone No.1 shall resume conversation. Phone No.2 shall not hear conversation.

n. Replace Info Turret switches to NORMAL and withdraw switchboard cords.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:18Z
Check the source to verify that this is the current version before use.

4.6.4.14 Intercommunication stations test.-

a. Area Intercom Tests-AN/MTA-4()

(1) Using a low resistance continuity tester measure continuity between PINS 26A and 26B of all 9 signal input connectors and BINDING POSTS marked AREA INTERCOM in Signal Interconnecting Box "A".

(2) Connect a TA-312()/PT phone to Area Intercom BINDING POSTS in Signal Interconnecting Box "A".

(3) Connect power cord plug of AREA INTERCOM at Wire Chief desk to receptacle in raceway. (Place Breaker #6 in Power Distribution Panel in ON position). Connect phone cord from AREA INTERCOM to corresponding phone jack on curbside wall.

(4) Turn Intercom to ON position. Pilot lamp shall light.

(5) Communications between phone and intercom shall be loud and clear.

(6) Turn Intercom to OFF position.

b. Local Intercom Tests

(1) Using a low resistance continuity tester measure continuity between BINDING POSTS marked LOCAL INTERCOM in Signal Entrance Box "A" on the AN/MTA-3() and AN/MTA-4().

(2) Connect power cord plugs of LOCAL INTERCOMS at the desk positions of the AN/MTA-3() and AN/MTA-4() to receptacles in raceway. (Place Breaker #6 in Power Distribution Panel in ON position). Connect phone cords from each LOCAL INTERCOM to respective phone jack at desk position.

(3) Turn Intercoms to ON position. Pilot lamp shall light in each case.

(4) Communication between Intercoms shall be loud and clear.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

(5) Turn Intercoms to OFF position.

4.6.4.15 Supervisor's phone test (Phones 1 and 2).-

a. Using a low resistance continuity tester measure continuity between BINDING POSTS marked PHONE 1 MTA-4 in Signal Interconnecting Box "A" on the AN/MTA-3() and BINDING POSTS marked PHONE 1 MTA-3 in Signal Interconnecting Box "A" on the AN/MTA-4().

b. Connect Line 98 to Line 99 on the Lineside of the Trunk MDF with jumpers.

c. Measure continuity between BINDING POSTS marked PHONE 2 Chief Operator in Signal Interconnecting Box "A" on AN/MTA-3() and BINDING POSTS marked PHONE 2 Wire Chief in Signal Interconnecting Box "A" on the AN/MTA-4().

d. In the AN/MTA-3(), connect a TA-312()/PT phone to the PHONE 1 MTA-4 jack with a phone cord.

e. In the AN/MTA-4(), connect a TA-312()/PT phone to the PHONE 1 MTA-3() jack with a phone cord.

f. Communication between phones shall be loud and clear.

g. Reconnect phones to PHONE 2 jacks in each assemblage. Communication between phones shall be loud and clear.

h. Remove jumpers from Lines 98 and 99 on Trunk MDF.

4.6.4.16 Spare phone jacks test.-

a. Connect a J-1077()/U to connector SIGNAL 9 in Signal Entrance Box "E" on the AN/MTA-4().

b. Using a low resistance continuity tester measure continuity between pairs 21 thru 25 on the J-1077()/U and the corresponding BINDING POSTS in Box "E"; and Lines 20 thru 24 on the Trunk MDF.

c. Connect TA-312()/PT phones to pair 21 in the J-1077()/PT and to SPARE jack 21 in the Jack Panel at the Wire Chief desk.

d. Communication between phones shall be loud and clear.

e. Repeat c. and d. above for pairs 22 thru 25.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:18Z
Check the source to verify that this is the current version before use.

4.7 Vehicular transportation test.- Each assemblage comprising the Central Office, Telephone Manual AN/MTC-1() shall be secured to a Government-furnished truck, 2-½ ton, M-35 and transported over the specially prepared course described in 4.7.1 for 10 laps at 5 mph, 10 laps at 10 mph, 10 laps at 15 mph, and 10 laps at 18 to 20 mph. A lap is defined as traversing the course in one direction. Each assemblage shall meet the requirements of paragraph 3.6 after completion of the test.

NOTE: Battery Dummy-Loads shall be installed for this test. (See 6.2(b)).

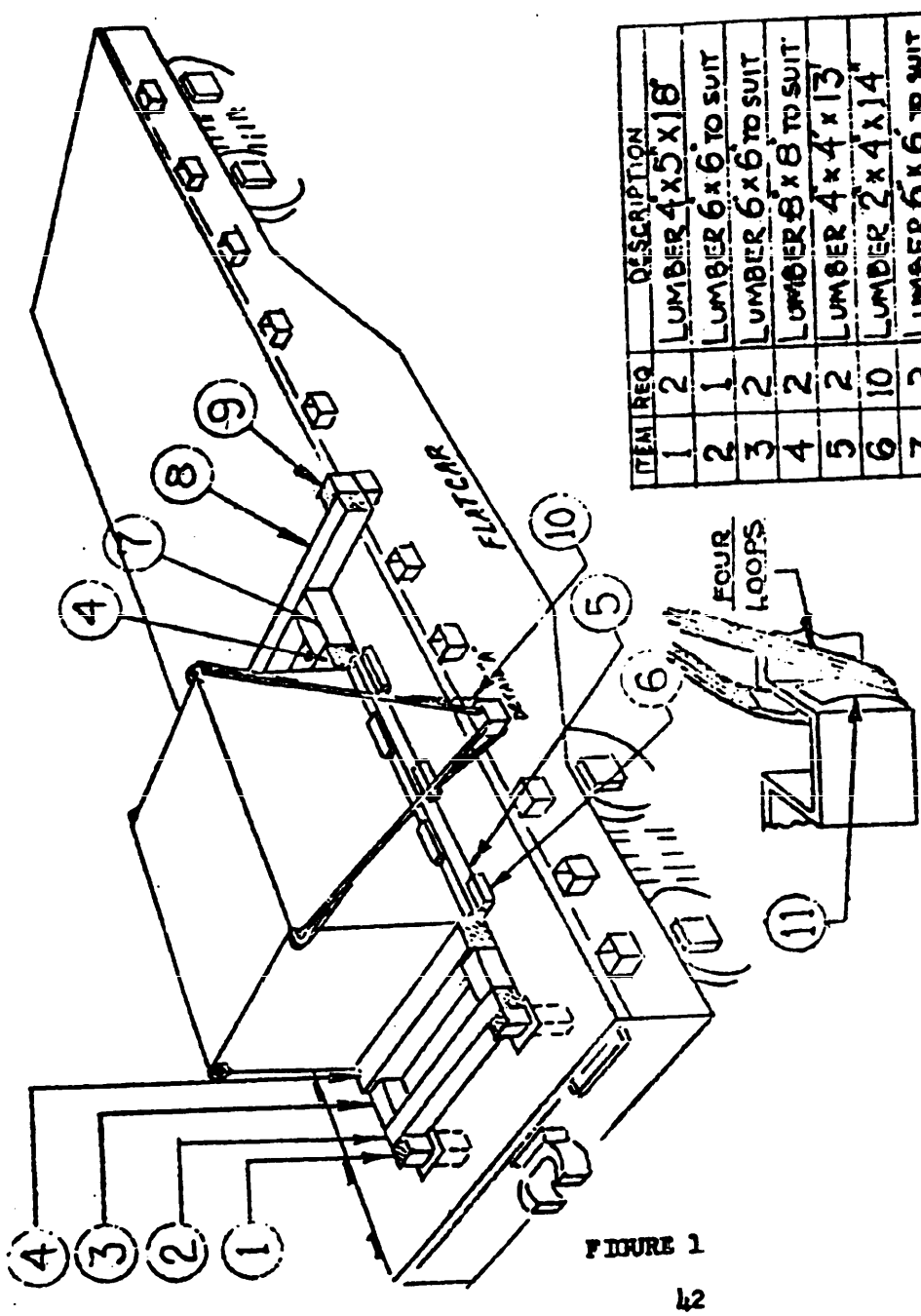
4.7.1 Vehicular transportation course.- The course for the military truck transportation test shall have twelve (12) 4-inch by 8-inch timbers placed 25 feet apart on the 8-inch face and with the 4-inch face fully above ground. The top edges shall have a 1 x 1-inch chamfer. The sixth and twelfth timbers shall be placed at 45 degrees to the direction of travel. The timbers shall be anchored securely by any convenient means. The ground on which the timbers are placed shall be hard and level.

4.8 Railroad transportation test.- Each assemblage comprising the AN/MTC-1() shall be loaded on a railroad flatcar (test car) in accordance with Figure 1. The blocking for each assemblage shall be independent of each other. The test shall be conducted on a flat stretch of track. A loaded car totaling 165,000 lbs traveling at 9 miles per hour (or equivalent impact energy conditions approved by the Government) shall be impacted against the test flatcar coupled to two other empty cars. These three cars shall be stationary with brakes off. Four impacts shall be performed, two impacts into each end of the test flatcar. The cabling and blocking holding the assemblages in place shall be tight at the start of each impact run. If the cabling or blocking is torn loose by the impact, the run shall be repeated. A damage survey shall be made after each impact and the AN/MTC-1() shall meet the requirements of 3.7.

NOTE: The tests shall be conducted without Batteries (or equivalent) installed.

4.9 Drop tests.- These tests shall be performed using Battery Dummy Loads. (See 6.2(b)).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.



ITEM	REQ	DESCRIPTION
1	2	LUMBER 4"x5"x18'
2	1	LUMBER 6"x6" TO SUIT
3	2	LUMBER 6"x6" TO SUIT
4	2	LUMBER 8"x8" TO SUIT
5	2	LUMBER 4"x4"x13'
6	10	LUMBER 2"x4"x14"
7	2	LUMBER 6"x6" TO SUIT
8	1	LUMBER 6"x6" TO SUIT
9	2	LUMBER TO SUIT
10	16	WIRE #8 SOLID, HARD, ZINC COATED
11	4	FLEX. METAL SHEETING

NOTES:
a. The blocking lumber shall be notched to accomodate towing eyes, bolts and rivet heads.
b. The blocking lumber shall be rigidly fastened to the floor of the flatcar.

FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.9.1 Flat drop.- Each assemblage comprising the Central Office, Telephone, Manual AN/MTC-1() shall be lifted and allowed to fall freely a distance of 18 inches with the skids impacting onto a hard concrete surface. This test shall be performed once. Upon completion of the test, each assemblage shall meet the requirements of 3.8.

4.9.2 Rotational drops.- Each assemblage comprising the Central Office, Telephone Manual, AN/MTC-1() shall have one edge placed on a nominal 4-inch high timber. The opposite edge of the assemblage shall be raised 18 inches and allowed to fall freely onto a hard concrete surface. This drop test shall be performed on each bottom edge of the assemblage. Upon completion of these drops. each assemblage shall meet the requirements of 3.8. In each drop test, the 4-inch high timber shall be placed under the skids and the eighteen inch measurement shall be taken from the bottom of the skids.

4.10 Watertightness.- Each assemblage comprising the AN/MTC-1(), without the use of any additional caulking, sealing, taping, etc., shall be subjected to the following tests:

a. Each of the five exposed surfaces of the shelters, with covers closed, shall be sprayed with water from nozzles, model G29SQ (or GG29SQ) as made by Spraying Systems Co., Bellwood, Illinois, or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each end panel shall be tested using nine nozzles, each side using fifteen nozzles, and the roof using six nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the roof nozzles which shall be at a distance of thirty-two inches. The row of nozzles in line with the air conditioner when testing the AN/MTA-3() shall be 19 inches measured from the extreme vertical surface of the air conditioner. (See Figure 2 for nozzle location). Each panel shall be sprayed for a period of twenty minutes. (If the watertightness test is to be followed by the fording test of 4.11, the bottom row of nozzles, with the exception of the nozzles facing the door end panel, can be eliminated). Upon completion of the above test, the AN/MTC-1() shall meet the requirements of 3.9. To determine that no water has entered space between the skins, holes shall be made at points to be determined by the Government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

b. Open all covers to operating position and repeat a. above.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

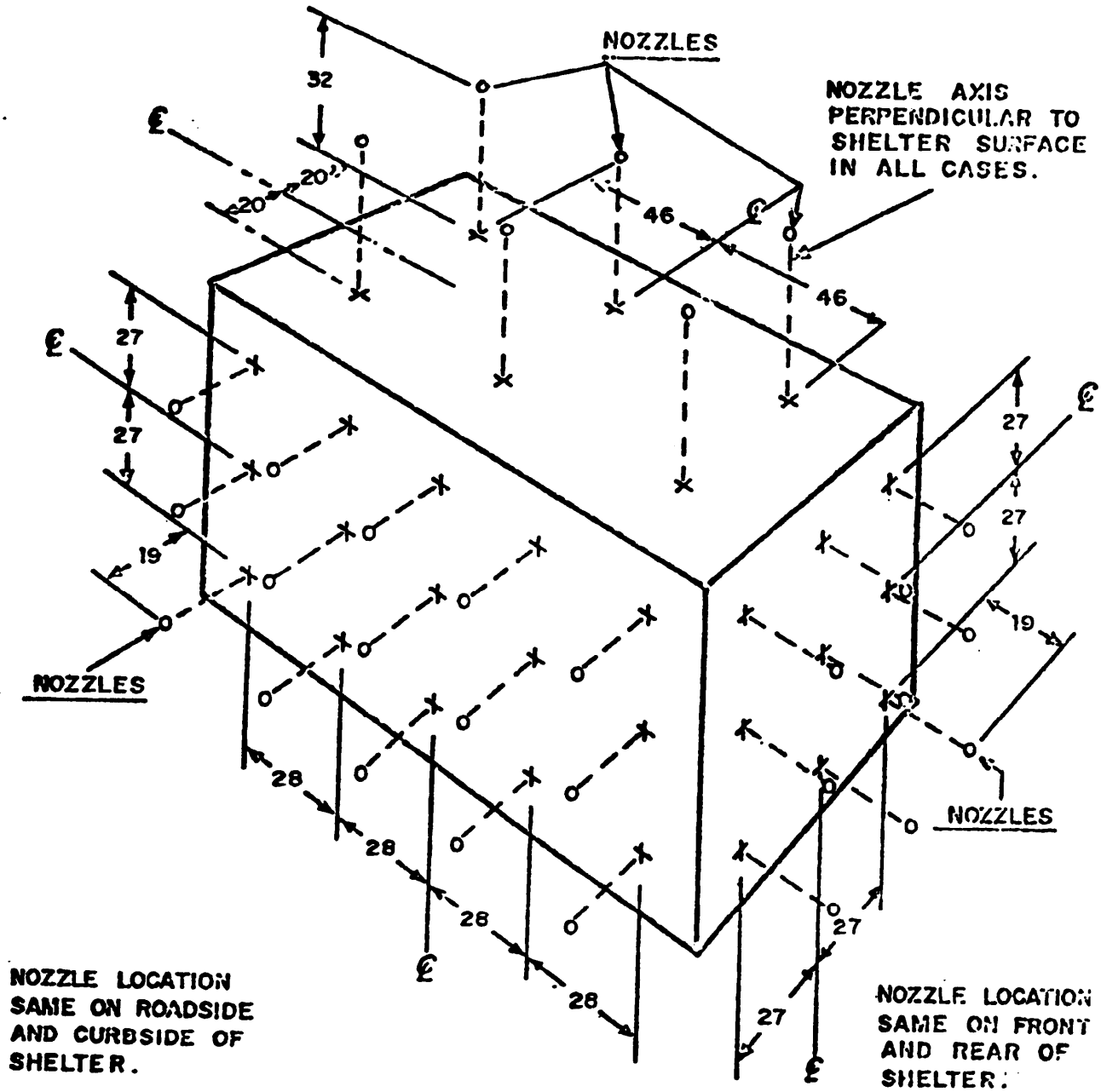


FIGURE 2

Note : GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.11 Fording test.- The shelters prepared for transportation, and without any additional sealer, caulking, taping, fording kit, etc, shall be immersed in water to a depth of 21 inches, measured from the bottom of the skids, for a period of one hour. The shelters shall be restrained by external tiedowns. Upon completion of this test, the shelters shall meet the requirements of 3.10. To determine that no water has entered space between the skins, holes shall be made at points to be determined by the government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

4.12 Mounting insert tests.-

4.12.1 Preproduction insert test.- Four of each size insert shall be randomly selected for test. Each insert selected shall be tested with an axial load followed by a torque load in accordance with Table V. After test the requirements of 3.11 shall be met. Failure of any insert shall be cause for rejection.

4.12.2 Production insert test.- Five of each size insert (if the total quantity of any size is less than 5, all of that size shall be tested) shall be randomly selected in each sample equipment. Each selected insert shall be tested in accordance with Table V. After test, the requirements of 3.11 shall be met. Failure of any insert shall be cause for rejection of the sample equipment.

Table V - Mounting Insert Loads

Insert Sizes	Preproduction Test Axial (lbs)	Preproduction Test Torque (Inch lbs)	Production Test Axial (lbs)	Production Test Torque (inch lbs)
1/4-20	1300	60	650	60
5/16-18	2000	100	1000	100

4.13 Testing of solderless terminal-lug connections.- Connections made with solderless terminal lugs shall be tested to meet the requirements of 3.21.4.

4.14 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.22. When a dimension is not within specified or design limits, it shall be considered a defect.

a. Size and mounting dimensions of brackets and mounting plates.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:18Z
Check the source to verify that this is the current version before use.

b. Size and mounting dimensions of racks and cabinets.

c. Size and mounting dimensions of equipment drawers.

d. Size, mounting dimensions, and capacity of ventilating equipment.

4.15 Visual and Mechanical inspection.- Equipment shall be examined for the defects listed in Standard MIL-STD-252, the requirements of 3.23 and Table VI.

Table VI - Classification of Visual and Mechanical Defects

Classification	Defects
Major	1. Mounting insert missing or mislocated. 2. Sealer improperly applied (holes, separations, or lack of adhesion). 3. Welds - cracked or porous. 4. Parts, brackets, or equipment loose or missing. 5. Drawers, doors, etc, inoperative. 6. Blower inoperative or rotating in wrong direction. 7. Mounting plates missing or mislocated. 8. Aircraft loading data plate, instructions or nameplate omitted, incorrect or illegible. 9. Power receptacles, switches, jacks, wire ducts or other electrical components loose, improperly located or inoperative. 10. Locking or holding devices missing or inoperative. 11. Lock washer or lock nut missing. 12. Burrs and sharp edges in wire duct not removed. 13. Incorrect or missing labels on binding posts and jacks.
Minor	1. Screw loose. 2. Doors, slides, drawers not easily operated. 3. Finish-abrasions or scratches. 4. Locking or holding devices difficult to operate. 5. Burrs and sharp edges on brackets not removed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.16 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with requirements of Section 5.

4.17 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling:

Paragraph 4.6.4	Systems operation
Paragraph 4.15	Visual and mechanical

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation, packaging, and packing of the AN/MTC-1( ) shall be in accordance with applicable requirement of Specification MIL-S-55505.

6. NOTES

6.1 Intended use.- The Central Office, Telephone, Manual AN/MTC-1( ) is intended for use by division and will provide one multiplied three-position switching facilities with termination for local and common battery subscriber circuits and manual and trunk dial circuits for a division area type communications system.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Preproduction and Group C Inspection of the AN/MTC-1( ) requires contractor fabricated dummy loads for the BB-46 Batteries (used in the vehicular transportation and drop tests) unless otherwise specified in the bid request or contract. The loads shall be configured to accommodate the battery hold-down bracketry.

c. A list of each equipment to be government loaned for testing the equipment.

d. A list of government furnished equipments required as part of the AN/MTC-1( ) on order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

e. Submission of the statement of treatment referenced in 3.19.4 as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.1 Examination.- Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.3.2 Testing.- Testing consists of determinations of compliance, using technical means.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.3).

6.5 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: AN/MTC-1K, AN/MTA-3M, AN/MTA-4M. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.6 Color.- The color chips referenced may be obtained upon request to Commanding General, Aberdeen Proving Grounds, ATTN: Coating and Chemical Laboratory, Aberdeen, Maryland 21005.

6.7 Verification inspection.- Verification by the government will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:
Army-EL

Preparing Activity:
Army - EL

Project No 5805-A125

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commanding General
US Army Electronics Command
Attn: AMSEL-TD-SS
Fort Monmouth, NJ 07703

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:18Z
Check the source to verify that this is the current version before use.